

Tan Dai Ngo

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EDUCATION

- University of Victoria** — Master of Engineering, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*
- Coursework: Algorithms and Data Models, Systems for Massive Datasets, Information Theory. GPA: 8.68 / 9.00.
- University of Chicago** — Master of Science, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*
- Coursework: Applied Software Engineering, Distributed Systems, Databases, Networks, Algorithms.
- University of Washington** — Bachelor of Science, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*
- Minors: Applied Mathematics, Informatics. GPA: 3.57 / 4.00.

EXPERIENCE

- Software Developer** — T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*
- Gathered goals from business stakeholders and wrote them up as specifications they could verify themselves.
 - Documented the root cause of each defect traced across service boundaries, so the fix outlived the ticket.
 - Built and maintained REST service endpoints in Java and Spring Boot inside a microservices codebase.
 - Reviewed teammates' code every sprint, catching defects before they reached release.
 - Cut manual effort on recurring dataset builds and validation runs by 80% by automating both in Python.
- Mathematics and Computer Science Teacher** — APU International Schools *Da Nang, Vietnam | Aug - Dec 2024*
- Explained technical material to 50+ students with no prior background, adjusting the pitch in real time.

PROJECTS

- Multi-Stage IoT Intrusion Detection** — Python, scikit-learn, XGBoost, PyTorch *Team Capstone, UVic*
- Compared six detection models under a 1% false-positive budget, reaching 92.5% recall at 0.96% FPR.
 - Designed a four-condition experiment to separate genuine learning from leakage in a public benchmark.
 - Proved the benchmark encodes capture-session identity, recoverable from flow features at ROC-AUC 0.922.
 - Reported the honest cost rather than the headline: PR-AUC falls 0.927 to 0.630 on a session-disjoint split.
- Ocean Data Platform with RAG** — Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project*
- Harmonized 10M+ sensor observations from NOAA and ONC sources into one layered, queryable model.
 - Evaluated retrieval over dataset metadata against a held-out query set: hit@k 1.0, term recall 0.875.
 - Forecast sensor readings with XGBoost on chronological holdouts rather than random splits.
 - Containerized every service so each experiment could be reproduced from scratch by someone else.
- Distributed Facility Reservation System** — Python, FastAPI, SQLite, pytest *Team Project, UChicago*
- Designed 27 versioned REST endpoints and published OpenAPI documentation for every one of them.
 - Wrote 71 pytest tests across the API, database, and reservation-rules layers, run on each change.
 - Negotiated one booking contract with four teams, then tested against their independently built services.
 - Secured every endpoint behind session tokens checked for freshness and permission scope.
- Anomaly Detection at Scale on Species Data** — Python, PySpark *Team Project, UVic*
- Implemented Isolation Forest from the published algorithm rather than a library, then validated the result.
 - Identified a previously undescribed variant of feature-specific swamping and proposed two mitigations.
 - Verified linear runtime growth on 1,093,203 records using stratified samples across eight quantile bins.

TECHNICAL SKILLS

Languages: Python, Java, SQL

AI Tooling: Claude Code for test authoring, debugging, refactoring, documentation, and plan validation

Web and APIs: REST API design, FastAPI, Spring Boot, OpenAPI documentation, session-based auth

Data and ML: pandas, NumPy, scikit-learn, XGBoost, PyTorch, Spark, PySpark, Qdrant vector store

Engineering Practice: Git, GitHub pull requests, code review, pytest, Docker, Jenkins, relational schema design